



Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LGD: Loss Given Default
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ SE: Standard Error
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive VaR and ES for the Normal and Student- t distributions and the sampling variance of their historical estimates
- ▣ Prove subadditivity of ES from the Rockafellar–Uryasev representation
- ▣ Estimate historical, filtered, extreme-value and CAViaR VaR/ES with honest intervals (HAC, block and residual bootstrap, declustering)
- ▣ Split portfolio risk into Euler contributions

- ▣ Background: Lecture 7; textbooks McNeil, Frey & Embrechts (2015) and Franke, Härdle & Hafner (2019)

Structure

- ▣ **Part A:** derivations (A1–A8)
- ▣ **Part B:** seven computational exercises with inference (B1–B7)
- ▣ **Part C:** one open question, a seed for the team project (C1), and a critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Sign convention: loss $L = -r$ in % of the position; VaR and ES are positive numbers
- Report every tail estimate with its uncertainty: how many observations lie beyond it?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Normal ES derivation, SE of historical VaR), A2 (Student- t ES derivation), A3 (a discrete loss), A5 (component VaR; (d) ES contributions)
- Part B: B1 (historical VaR and ES on BET; bootstrap, asymptotic and HAC intervals), B2 (GPD fit on BET; extremal index)

Proposed: you solve them

- Part A: A4 (failure of subadditivity; model: A3), A6 (validity of Cornish–Fisher; model: A1), A7 (ES as a minimum; model: A3, A4), A8 (variance of historical ES; model: A1)
- Part B: B3 (filtered vs historical on Bitcoin; model: B1), B4 (Cornish–Fisher vs empirical; model: A1, B1), B5 (component VaR of a portfolio; model: A5), B6 (bootstrap interval for ES; model: B1), B7 (CAViaR on BET; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: B1)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter7_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: BET (2000–2026), S&P 500, Bitcoin (2014–2026), EUR/RON (BNR reference rate), gold; SPY, TLT, GLD; eight BVB blue chips; daily closes up to 18 September 2026
- Equity indices: weekdays only; holiday records (unchanged close, zero volume) removed, genuine unchanged closes kept as zero returns; crypto keeps all 7 days of the week
- Portfolios: prices joined on common trading days first, then simple returns

Formulas Used Today

- Level α = tail probability (VaR 1%, ES 2.5%); return X , loss $L = -X$: $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$,
 $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$
- Discrete case [Acerbi & Tasche \(2002\)](#): $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$
- Normal distribution, $z_\alpha = \Phi^{-1}(\alpha) < 0$: $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$, $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$
- Student- t : $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{s\nu(t_\alpha)}{\alpha} \frac{\nu+t_\alpha^2}{\nu-1}$; variance σ^2 when $s = \sigma\sqrt{(\nu-2)/\nu}$
- GPD tail of the losses: $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} \alpha \right)^{-\xi} - 1 \right]$, $\text{ES}_\alpha = \frac{\text{VaR}_{\alpha+\beta-\xi u}}{1-\xi}$ ($\alpha < N_u/n$; ES needs $\xi < 1$)
- Euler: $C_i = w_i \partial \text{VaR} / \partial w_i = \mathbf{z}_{1-\alpha} w_i (\Sigma w)_i / \sigma_p$, $\sum_i C_i = \text{VaR}$ [Tasche \(2008\)](#)

A1: Normal VaR and ES [Solved]: Task

□ Data and assumptions

- ▶ Daily returns of a position are Normal, $X \sim N(\mu, \sigma^2)$, with $\mu = 0.04\%$ and $\sigma = 1.2\%$; the loss is $L = -X$.
- ▶ We write $z_\alpha = \Phi^{-1}(\alpha) < 0$, and Φ , φ for the standard Normal distribution and density functions.

□ Tasks

- ▶ (a) Derive $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$ starting from the definition $ES_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$.
- ▶ (b) Compute VaR and ES at $\alpha = 1\%$ and at $\alpha = 2.5\%$, in % of the position.
- ▶ (c) Compute the asymptotic standard error of the *historical* VaR 1%, $SE = \sqrt{\alpha(1-\alpha)/n}/f(\text{VaR}_\alpha)$, for $n = 500$ and for $n = 6\,670$ days, and the width of the 95% interval.

□ Report

- ▶ The four numbers of (b), the two standard errors of (c), and a comparison with the bootstrap interval widths of B1: why are the B1 intervals wider?

A1: Normal VaR and ES [Solved]: Solution (1/2)

- **(a) Derivation.** Write $X = \mu + \sigma Z$; the event $X \leq q_\alpha$ is $Z \leq z_\alpha$, which has probability α .
 - ▶ Since $\varphi'(u) = -u\varphi(u)$, we get $\int_{-\infty}^z u\varphi(u) du = [-\varphi(u)]_{-\infty}^z = -\varphi(z)$.
 - ▶ Hence $\mathbb{E}[X \mid X \leq q_\alpha] = \mu + \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = \mu - \sigma \frac{\varphi(z_\alpha)}{\alpha}$, and changing the sign gives the formula.
- **(b) Numbers at 1%.** The quantile is $z_{1\%} = -2.326$ and the density there is $\varphi(z_{1\%}) = 0.0267$, so $\varphi/\alpha = 2.665$.
 - ▶ $\text{VaR}_{1\%} = -(0.04 + 1.2 \cdot (-2.326)) = 2.75\%$ and $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16\%$.
- **(b) Numbers at 2.5%.** Here $z_{2.5\%} = -1.960$ and $\varphi/\alpha = 2.338$.
 - ▶ $\text{VaR}_{2.5\%} = 2.31\%$ and $\text{ES}_{2.5\%} = -0.04 + 1.2 \cdot 2.338 = 2.77\%$.
 - ▶ Interpretation: ES 2.5% (2.77%) is almost equal to VaR 1% (2.75%); this near-equality under the Normal distribution is why FRTB chose ES 2.5%.

A1: Normal VaR and ES [Solved]: Solution (2/2)

- (c) **Density at the quantile.** The loss density at VaR 1% is $f(\text{VaR}_{1\%}) = \varphi(z_{1\%})/\sigma = 0.0267/1.2 = 0.0222$.
 - ▶ $n = 500$: $\text{SE} = \sqrt{0.01 \cdot 0.99/500}/0.0222 = 0.20$ percentage points, so the 95% interval is 0.79 points wide.
 - ▶ $n = 6\,670$: $\text{SE} = 0.055$ and the 95% width is 0.22 points.
- **Comparison with B1.** The bootstrap intervals for BET VaR 1% are 1.01 points wide (500 days) and 0.61 points wide (full sample).
 - ▶ Interpretation: BET losses have a heavier tail than the Normal distribution, so the density at VaR is lower and the quantile is estimated less precisely.
 - ▶ The standard error falls like $1/\sqrt{n}$: thirteen times more data make the interval about $\sqrt{13.3} \approx 3.7$ times narrower.

A2: Student- t ES [Solved]: Task

□ Data and assumptions

- ▶ Returns are $X = m + sT$ with T Student- t with ν degrees of freedom and density $g_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2}$.
- ▶ The quantile is $t_\alpha = t_\nu^{-1}(\alpha) < 0$; for the numbers use $\nu = 4$, $m = 0.04\%$ and the same standard deviation 1.2% as in A1.

□ Tasks

- ▶ (a) Show that $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu + t^2}{\nu - 1}$ for $\nu > 1$, and deduce $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$.
- ▶ (b) Find the scale s that gives a standard deviation of 1.2%, using $\text{Var}(T) = \nu/(\nu - 2)$.
- ▶ (c) Compute VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$.

□ Report

- ▶ The four numbers of (c) next to those of A1; interpretation question: at which level do the two models differ most, and why?

A2: Student- t ES [Solved]: Solution

- (a) **Antiderivative.** Differentiate $(1 + x^2/\nu)^{-(\nu-1)/2}$: the result is $-\frac{\nu-1}{\nu} x (1 + x^2/\nu)^{-(\nu+1)/2}$.
 - Therefore $xg_\nu(x) = -\frac{\nu c_\nu}{\nu-1} \frac{d}{dx} (1 + x^2/\nu)^{-(\nu-1)/2}$, and integrating from $-\infty$ to t gives $-\frac{\nu}{\nu-1} g_\nu(t)(1 + t^2/\nu)$.
 - This equals $-g_\nu(t)(\nu + t^2)/(\nu-1)$; the limit at $-\infty$ vanishes only if $\nu > 1$. Divide by α and rescale by $m + sT$ as in A1.
- (b) **Scale.** $s^2\nu/(\nu-2) = 1.2^2$, so $s = 1.2\sqrt{2/4} = 0.849$.
- (c) **At 1%.** $t_{1\%} = -3.747$ and $g_4(t_{1\%}) = 0.0087$, so the ES factor is $\frac{0.0087}{0.01} \cdot \frac{4 + 14.04}{3} = 5.221$.
 - $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14\%$ and $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39\%$.
- (c) **At 2.5%.** $t_{2.5\%} = -2.776$ gives $\text{VaR}_{2.5\%} = 2.32\%$ and $\text{ES}_{2.5\%} = 3.35\%$.
 - Interpretation: VaR 2.5% is almost the same in both models (2.32 vs 2.31), while VaR 1% and both ES values are much higher under t_4 : heavy tails show only deep in the tail, and ES sees them first.

A3: VaR and ES of a Discrete Loss [Solved]: Task

□ Data and assumptions

- ▶ A bond loses 60 (loss given default, LGD) with probability $p = 3\%$ and 0 otherwise, so the return is $X = -60$ or $X = 0$.
- ▶ Use the general definitions $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$ and the discrete ES formula of the lecture [Acerbi & Tasche \(2002\)](#).

□ Tasks

- ▶ (a) Compute VaR 5% and VaR 2.5% directly from the definition.
- ▶ (b) Compute ES 5% and ES 2.5% with $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$.
- ▶ (c) Explain why $\text{ES}_{5\%}$ is not equal to $\mathbb{E}[L \mid L \geq \text{VaR}_{5\%}]$.

□ Report

- ▶ The four numbers and a one-sentence answer to (c).

A3: VaR and ES of a Discrete Loss [Solved]: Solution

- (a) **VaR 5%.** $P(X \leq -60) = 3\%$ is not above 5%, while $P(X \leq 0) = 100\% > 5\%$; so $q_{5\%} = 0$ and $\text{VaR}_{5\%} = 0$.
 - ▶ VaR 2.5%: $P(X \leq -60) = 3\% > 2.5\%$, so $q_{2.5\%} = -60$ and $\text{VaR}_{2.5\%} = 60$.
- (b) **ES 5%.** $\mathbb{E}[L1\{L > 0\}] = 60 \cdot 0.03 = 1.8$ and $P(L > 0) = 3\%$, so $\text{ES}_{5\%} = \frac{1.8 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$.
 - ▶ ES 2.5%: nothing lies above 60, so $\text{ES}_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$.
- (c) **Why the correction term.** $\mathbb{E}[L \mid L \geq 0] = 1.8$ averages over **all** days, not over the worst 5%.
 - ▶ Interpretation: with atoms, the worst 5% of outcomes consist of the 3% default mass plus 2% of the zero-loss mass; the formula weights them exactly, and ES then equals 36.

A4: When VaR Punishes Diversification [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Two independent bonds A and B; each loses 100 with probability $p = 0.7\%$ and 0 otherwise [Artzner et al. \(1999\)](#). The portfolio is $A + B$.

□ Tasks

- ▶ (a) Write the distribution of the portfolio loss (values 0, 100, 200 and their probabilities).
- ▶ (b) Compute VaR 1% and ES 1% of one bond and of the portfolio, with the discrete formula of A3.
- ▶ (c) Show that VaR 1% is not subadditive here while ES 1% is.
- ▶ (d) Repeat (b)–(c) at $\alpha = 1.5\%$ and find the range of α for which VaR fails.

□ Report

- ▶ A small table of VaR and ES for A, B and $A + B$ at both levels, and one sentence on what a VaR-based limit would reward.
- ▶ Model: A3.

A5: Component VaR and ES of Two Assets [Solved]: Task

□ Data and assumptions

- ▶ Equity 600 000 EUR with $\sigma_1 = 1.2\%$ daily; bonds 400 000 EUR with $\sigma_2 = 0.9\%$; correlation $\rho = -0.2$; Normal distribution, zero mean, $\alpha = 1\%$.

□ Tasks

- ▶ (a) Compute the portfolio standard deviation in EUR and the portfolio VaR 1%.
- ▶ (b) Compute the marginal VaR $z_{1-\alpha}(\Sigma w)_i/\sigma_p$ and the component VaR C_i of each asset, and check $C_1 + C_2 = \text{VaR}$ Tasche (2008).
- ▶ (c) Compute the diversification benefit (sum of stand-alone VaRs minus portfolio VaR).
- ▶ (d) Using $\text{ES}_\alpha(w) = \min_v \{v + \mathbb{E}[(L_p - v)_+]/\alpha\}$ Rockafellar & Uryasev (2000), show that $w_i \partial \text{ES}_\alpha / \partial w_i = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha]$ and evaluate these ES contributions here.

□ Report

- ▶ A table of C_i and C_i^{ES} with shares; interpretation: do the bonds carry risk in proportion to their money weight?

A5: Component VaR and ES of Two Assets [Solved]: Solution (1/2)

- (a) **Portfolio risk.** Stand-alone standard deviations: $600\,000 \cdot 1.2\% = 7\,200$ EUR and $400\,000 \cdot 0.9\% = 3\,600$ EUR.
 - ▶ $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$, so $\sigma_p = 7\,378$ EUR and $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$ EUR.
- (b) **Marginal and component VaR.** $(\Sigma w) = (77.76, 19.44)$ (in EUR, with Σ in decimal units), which gives the marginal VaR per EUR of exposure $(0.0245, 0.0061)$.
 - ▶ $C_1 = 14\,711$ EUR (85.7%) and $C_2 = 2\,452$ EUR (14.3%); their sum is 17 163 EUR, exactly the portfolio VaR.
- (c) **Diversification.** Stand-alone VaRs $16\,750 + 8\,375 = 25\,125$ EUR against a portfolio VaR of 17 163 EUR.
 - ▶ Interpretation: the bonds hold 40% of the money but only 14.3% of the risk, because the negative correlation makes them a partial hedge.

A5: Component VaR and ES of Two Assets [Solved]: Solution (2/2)

- **(d) Envelope argument.** $v^* = \text{VaR}_\alpha$ minimises $G(w, v) = v + \mathbb{E}[(L_p - v)_+]/\alpha$, so $\partial \text{ES}_\alpha / \partial w_i = \partial G / \partial w_i$ at v^* .
 - ▶ Differentiating under the expectation: $\partial G / \partial w_i = \mathbb{E}[-R_i \mathbf{1}\{L_p > v^*\}]/\alpha$.
 - ▶ For a continuous L_p , $P(L_p > \text{VaR}_\alpha) = \alpha$, so this is $\mathbb{E}[-R_i \mid L_p \geq \text{VaR}_\alpha]$; multiplying by w_i and summing gives $\mathbb{E}[L_p \mid L_p \geq \text{VaR}_\alpha] = \text{ES}_\alpha$.
- **(d) Numbers.** Under the Normal distribution $\text{ES}_\alpha = \sigma_p \varphi(z_\alpha) / \alpha$, so every contribution scales by $\frac{\varphi(z_\alpha) / \alpha}{z_{1-\alpha}} = 1.146$.
 - ▶ $C_1^{\text{ES}} = 16\,854$ EUR and $C_2^{\text{ES}} = 2\,809$ EUR; together 19 663 EUR, the portfolio ES 1%.
 - ▶ Interpretation: under the Normal distribution the ES shares equal the VaR shares, so ES adds no new allocation information; with heavy tails it does (see B5).

A6: When Is Cornish–Fisher a Valid Quantile? [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Returns with $\mu = 0$, $\sigma = 1.2\%$, skewness $S = -0.5$ and excess kurtosis K [Cornish & Fisher \(1938\)](#).
- ▶ The Cornish–Fisher map is $\tilde{z}(z) = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$ and $\text{VaR}_\alpha = -\sigma\tilde{z}(z_\alpha)$.

□ Tasks

- ▶ (a) With $K = 3$, compute the three correction terms at $z_{1\%}$, then $\tilde{z}_{1\%}$ and the Cornish–Fisher VaR 1%; compare with the Normal VaR.
- ▶ (b) Derive $d\tilde{z}/dz$ and find where it is smallest on $[z_{0.1\%}, 0]$.
- ▶ (c) Find the largest K for which $\tilde{z}$ is increasing on this interval, and decide whether $K = 10$ (close to the S&P 500) is admissible.

□ Report

- ▶ The VaR of (a), the bound on K , and whether a VaR computed with $K = 10$ is a valid estimate. Model: A1.

A7: ES as a Minimum, and Subadditivity [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ L is any integrable loss; define $F_L(v) = v + \mathbb{E}[(L - v)_+]/\alpha$ Rockafellar & Uryasev (2000). For the numbers, use the A4 portfolio (two bonds, $p = 0.7\%$) at $\alpha = 1\%$.

□ Tasks

- ▶ (a) Prove that F_L is convex and that its minimum over v equals $\text{ES}_\alpha(L)$, attained at $v = \text{VaR}_\alpha(L)$.
- ▶ (b) Deduce $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$ from $(a + b)_+ \leq a_+ + b_+$.
- ▶ (c) For the A4 portfolio, compute the slope of F on $(0, 100)$ and on $(100, 200)$, the minimiser and the minimum.

□ Report

- ▶ The two proofs in a few lines and the three numbers of (c), compared with $\text{ES}_{1\%}(A + B)$ from A4. Model: A3, A4.

A8: The Sampling Variance of Historical ES [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ n i.i.d. losses with a continuous distribution, a positive density at VaR_α and $\mathbb{E}[L^2] < \infty$; historical ES is the mean of the losses beyond the empirical VaR. Its influence function is $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+ / \alpha - \text{ES}_\alpha$ [Chen \(2008\)](#).

□ Tasks

- ▶ (a) Show that the variance of the limit of $\sqrt{n}(\widehat{\text{ES}}_\alpha - \text{ES}_\alpha)$ is $\text{avar}(\widehat{\text{ES}}_\alpha) = \text{Var}(\text{IF}(L)) = \text{Var}((L - \text{VaR}_\alpha)_+) / \alpha^2$, and explain why estimating VaR adds no term.
- ▶ (b) For $L \sim N(0, 1)$ and $\alpha = 2.5\%$, compute $\mathbb{E}[(L - v)_+]$ and $\mathbb{E}[(L - v)_+^2]$ in closed form at $v = \text{VaR}_\alpha$.
- ▶ (c) Compute the standard error of $\widehat{\text{ES}}_{2.5\%}$ for $n = 500$, for $\sigma = 1$ and for $\sigma = 1.2\%$, and compare with the S&P 500 over its last 500 days.

□ Report

- ▶ The two moments, the asymptotic variance and the two standard errors; interpretation: why is the S&P 500 value much larger? Model: A1.

B1: Historical VaR and ES on BET [Solved]: Task

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ Two samples: the full period ($n = 6\,670$) and the last 500 days (from 2024-09-12).

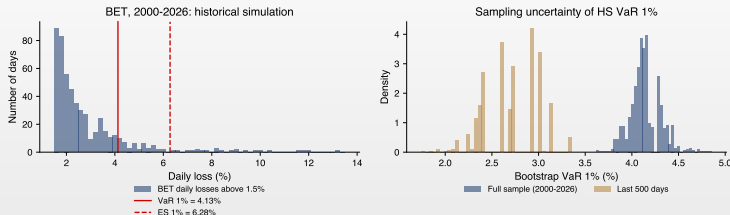
□ Tasks

- ▶ (a) Compute historical VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$ on both samples (VaR = empirical $(1 - \alpha)$ quantile of L , ES = mean of the losses at or above it).
- ▶ (b) Build a 95% percentile bootstrap interval for VaR 1% with 2000 i.i.d. resamples [Efron \(1979\)](#).
- ▶ (c) Build three more 95% intervals for VaR 1%: asymptotic i.i.d. ($\widehat{\text{VaR}} \pm 1.96 \sqrt{\alpha(1 - \alpha)/n/\hat{f}}$, Gaussian kernel density, Silverman bandwidth), asymptotic HAC (Bartlett kernel, lag $\lfloor 4(n/100)^{2/9} \rfloor$), and a moving-block bootstrap with 20-day blocks [Künsch \(1989\)](#).

□ Report

- ▶ A table with the eight VaR/ES numbers and the four intervals. Interpretation: how many losses lie beyond VaR 1% in each sample, which estimate would you report to a risk committee, and which interval is widest and why?

B1: Historical VaR and ES on BET [Solved]: Solution (1/3)



	n	$\text{VaR}_{1\%}$	$\text{ES}_{1\%}$	$\text{VaR}_{2.5\%}$	$\text{ES}_{2.5\%}$
2000–2026	6 670	4.13	6.28	2.80	4.57
Last 500 days	500	2.71	3.22	2.05	2.69

- (a) **Procedure.** Sort the losses, take the 99% (97.5%) empirical quantile as VaR and average the losses at or above it for ES; the chart shows the tail and the bootstrap distributions of VaR 1%.

B1: Historical VaR and ES on BET [Solved]: Solution (2/3)

- (a) **Tail counts.** Beyond VaR 1% lie 67 losses in the full sample but only 5 in the last 500 days.
 - ▶ So the 500-day ES 1% is the average of only five numbers and should be reported with a warning.
- (b) **Bootstrap intervals.** Full sample [3.85, 4.46]; last 500 days [2.11, 3.12].
 - ▶ The two intervals do not overlap: the recent market is calmer than the 26-year average.
 - ▶ Interpretation: report the recent value for today's risk and the long-run value as a stress reference.

B1: Historical VaR and ES on BET [Solved]: Solution (3/3)

	i.i.d. bootstrap	asympt., i.i.d.	asympt., HAC	block bootstrap
2000–2026	[3.85, 4.46]	[3.81, 4.44]	[3.69, 4.57]	[3.61, 4.80]
Last 500 days	[2.11, 3.12]	[2.18, 3.25]	[2.19, 3.23]	[2.02, 3.03]

- **(c) Asymptotic step.** $\hat{f}(\widehat{\text{VaR}}) = 0.0076$ (full) and 0.0163 (500 days), so $\text{SE} = \sqrt{0.01 \cdot 0.99/n/\hat{f}} = 0.16$ and 0.27 ; the HAC standard errors are 0.23 and 0.27 .
- **Reading the table.** Full-sample widths: $0.61, 0.63, 0.88, 1.19$; exceedances cluster in 2008 and 2020, so HAC and blocks widen the interval by a half to a factor of two.
- Last 500 days: all four widths are about 1; the period is calm, with little clustering, so the i.i.d. answer is adequate there.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Task

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ Peaks over threshold: the excesses $L - u$ above a high threshold u follow a generalised Pareto distribution (GPD) with shape ξ and scale β [McNeil, Frey & Embrechts \(2015\)](#).

□ Tasks

- ▶ (a) Set u so that the largest 5% of losses lie above it and fit the GPD to the excesses by maximum likelihood.
- ▶ (b) Compute the i.i.d. standard errors $SE(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$ and $SE(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$.
- ▶ (c) Compute the GPD VaR and ES at $\alpha = 1\%$, 0.5% and 0.1% and compare with historical simulation.
- ▶ (d) Refit for thresholds leaving from 10% down to 1.5% of the losses above them and plot $\hat{\xi}$ with 95% bands.
- ▶ (e) Estimate the extremal index θ of the exceedances of u with the intervals estimator [Ferro & Segers \(2003\)](#), decluster, and refit the GPD on the cluster maxima.

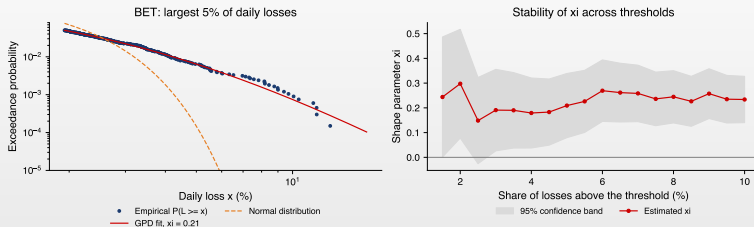
□ Report

- ▶ The fit, the VaR/ES table, the stability chart and the declustered fit. Interpretation: is the BET tail heavier than the Normal distribution, how sure are we, and is the i.i.d. SE of $\hat{\xi}$ credible?

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (1/3)

- (a) **Fit.** $u = 1.93\%$ leaves $N_u = 334$ of $n = 6\,670$ losses above it; maximum likelihood gives $\hat{\xi} = 0.209$ and $\hat{\beta} = 1.20$.
- (b) **Standard errors.** $SE(\hat{\xi}) = (1 + 0.209)/\sqrt{334} = 0.066$ and $SE(\hat{\beta}) = 0.10$.
 - ▶ $\hat{\xi}/SE = 3.2$: under independence the tail is significantly heavier than exponential (Normal-type); (e) checks the independence assumption.
- (c) **VaR 1%, step by step.** $N_u/n = 0.050$, so $(0.01/0.050)^{-0.209} = 1.401$.
 - ▶ $VaR_{1\%} = 1.93 + \frac{1.20}{0.209}(1.401 - 1) = 4.22\%$ (historical 4.13%); $ES_{1\%} = 6.34\%$ (historical 6.28%).
 - ▶ Deeper: VaR 0.5% 5.47 vs 5.38; VaR 0.1% 9.18 vs 9.50, with only 7 observations beyond; ES 0.1% 12.61 vs 11.25.
 - ▶ Interpretation: the two methods agree down to 0.5%; at 0.1% historical ES averages seven numbers, and only the GPD gives a stable value.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (2/3)



- (d) **Stability.** Across thresholds $\hat{\xi}$ ranges from 0.15 to 0.30, but VaR 0.1% only from 9.04% to 9.51%.
- Reading the chart: left, the GPD follows the empirical tail and the Normal tail falls far below it; right, the band widens as fewer losses remain. The quantile is more stable than the shape parameter.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (3/3)

- (e) **Extremal index.** From the gaps T_i between the 334 exceedances: $\hat{\theta} = \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} = 0.32$.
- ▶ Declustering [Ferro & Segers \(2003\)](#): target $C = \lfloor \hat{\theta}N \rfloor + 1 = 108$ clusters, cut at the $C - 1$ largest gaps; the $(C - 1)$ -th and C -th largest gaps are equal, so C is reduced until they differ, leaving 100 clusters of 3.3 exceedances on average.
 - ▶ At the VaR 1% threshold $\hat{\theta} = 0.40$.
- **Declustered fit.** GPD on the 100 cluster maxima: $\hat{\xi} = 0.269$ (SE 0.127) against 0.209 (SE 0.066) on all exceedances.
- ▶ $\hat{\xi}/\text{SE}$ falls from 3.2 to 2.1: only about 100 of the 334 excesses carry independent information.
 - ▶ Interpretation: the tail is still heavy, but the i.i.d. SE of 0.066 is not credible; report the declustered fit, or filter with GARCH first (conditional EVT).

B3: Filtered vs Historical Simulation on Bitcoin [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Bitcoin in USD, data/market/BTC-USD.CC.csv, daily closes 17 September 2014 – 18 September 2026, 7 days a week; daily log returns in %.
- ▶ Model: $r_t = \mu_t + \sigma_t z_t$ with AR(1) mean and GARCH(1,1) variance, estimated by quasi-maximum likelihood Barone-Adesi et al. (1999).

□ Tasks

- ▶ (a) Estimate the model on the full sample and forecast $\hat{\mu}_{t+1}$ and $\hat{\sigma}_{t+1}$ for 19 September 2026.
- ▶ (b) Compute the filtered VaR 1% and ES 1% from the empirical quantile of the standardised residuals, and compare with historical simulation on 500 days and on the full sample.
- ▶ (c) Using one-day-ahead forecasts (GARCH re-estimated each January), count the days of the last two years with a loss above each VaR.
- ▶ (d) Build a 90% residual-bootstrap interval for today's filtered VaR 1% with 999 paths, re-estimating the GARCH model on each path Christoffersen & Gonçalves (2005).

□ Report

- ▶ The three VaR/ES pairs, the exceedance counts and the interval. Interpretation: why can the three VaRs differ by a factor of two, which method misses more often, and does the interval of (d) overlap the 500-day historical VaR? Model: B1.

B4: Cornish–Fisher vs Empirical Quantiles [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Daily log returns in % (data/market): S&P 500 and gold 2000–2026, BET 2000–2026, Bitcoin 2014–2026, EUR/RON (BNR reference rate) 2005–2026; full samples.

□ Tasks

- ▶ (a) Compute the sample skewness S and excess kurtosis K of the returns of each series.
- ▶ (b) Compute VaR 1% and VaR 2.5% three ways: Normal, Cornish–Fisher (formula of A6) and historical.
- ▶ (c) Build a 95% percentile bootstrap interval (1 000 i.i.d. resamples) for the historical VaR 1% and check whether the Cornish–Fisher value lies inside.

□ Report

- ▶ One table with all numbers. Interpretation: is Cornish–Fisher inside the interval for any series, why does it work better at 2.5% than at 1%, and would moments from the last year help? Model: A1, A6, B1.

B5: Component VaR of a Four-Asset Portfolio [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ SPY, TLT, GLD (adjusted closes) and Bitcoin, data/market, 18 September 2014 – 18 September 2026; prices joined on the 3017 common days first, then daily simple returns in %.
- ▶ Equal weights of 25%, rebalanced daily; $\alpha = 1\%$ for VaR and $\alpha = 2.5\%$ for ES.

□ Tasks

- ▶ (a) Compute the Normal VaR 1% $z_{1-\alpha} \sqrt{w^\top \Sigma w}$, the marginal VaR and the component VaR of each asset [Tasche \(2008\)](#).
- ▶ (b) Compute the historical ES 2.5% of the portfolio and the contributions $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$ as averages over the tail days.
- ▶ (c) Compute the undiversified VaR $\sum_i z_{1-\alpha} w_i \sigma_i$.

□ Report

- ▶ A table of contributions and shares. Interpretation: is an equally weighted portfolio equally risky in each asset, how does it compare with the 40/30/20/10 portfolio of the lecture, and which weights would equalise the contributions? Model: A5.

B6: How Precise Is an ES Estimate? [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ S&P 500 (data/market/GSPC.INDX.csv) and BET, daily log returns in %, 2000–2026; two samples each: full and last 500 days.

□ Tasks

- ▶ (a) Compute historical ES 2.5% (lecture convention), the tail count $n\alpha$ and the number k of losses the tail mean actually averages.
- ▶ (b) Build a 95% percentile interval with the i.i.d. bootstrap [Efron \(1979\)](#), 2 000 resamples.
- ▶ (c) Repeat (b) with the moving-block bootstrap, blocks of 20 days [Künsch \(1989\)](#).

□ Report

- ▶ A table with $n\alpha$, ES and both intervals. Interpretation: why is the block interval wider on the full sample, and does a 500-day unconditional ES estimate pin ES down to within 0.3 percentage points [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)? Model: B1.

B7: CAViaR on BET [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ CAViaR with symmetric absolute value: $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$, $\alpha = 1\%$, $\text{VaR}_1 = -\hat{q}_\alpha$ of the first 300 returns (their upper loss quantile), these days kept in the estimation sample [Engle & Manganeli \(2004\)](#).

□ Tasks

- ▶ (a) Estimate β on 2000–2015 by minimising the pinball loss $\sum_t (\alpha - 1\{r_t < -\text{VaR}_t\})(r_t + \text{VaR}_t)$: 10^4 random starts in $(0, 1)^3$, the best 10 refined by simplex and quasi-Newton steps.
- ▶ (b) With β fixed, forecast VaR for 2016–2026 and compute the exceedance rate and the mean pinball loss; compare with FHS (GARCH re-estimated each January) and 500-day historical simulation.
- ▶ (c) For 20 February – 30 April 2020 count the exceedances and report the peak VaR of each method.

□ Report

- ▶ The estimates, a comparison table and the COVID-19 numbers. Interpretation: what does $\hat{\beta}_1$ say about the memory of BET risk, how does CAViaR react to the crash compared with FHS, and are the exceedance counts different? (formal tests in Chapter 8). Model: B3; lecture, CAViaR on the S&P 500.

C1: ES Capital, BVB vs S&P 500 [Open]: Task

□ Data and assumptions

- ▶ BVB basket: Banca Transilvania, OMV Petrom, BRD, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica (data/market, adjusted closes), equal weights, rebalanced daily; S&P 500 portfolio: SPY.
- ▶ Last ten years (from 2016-09-20), daily simple returns R_t in % in local currency, 1 million invested; a simplified illustration inspired by FRTB [Basel Committee \(2019\)](#), not the full capital calculation of [Basel Committee, MAR33](#).

□ Tasks

- ▶ (a) Compute the one-day historical ES 2.5% of each portfolio (loss $-R_t$).
- ▶ (b) Compute the 10-day ES 2.5% three ways: $\sqrt{10}$ scaling; overlapping compounded losses $100 [1 - \prod_{i=1}^{10} (1 + R_{t+i}/100)]$; filtered Monte Carlo from today's AR(1)-GARCH(1,1) state on the log returns, each path converted to a simple loss.
- ▶ (c) Stress window: the 250 consecutive days with the largest one-day historical ES 2.5%; compute the stressed ES inside it at 1 day and, by $\sqrt{10}$ scaling, at 10 days.
- ▶ (d) Capital per million = $m_c \times$ stressed ES, $m_c = 1.5$, for a liquidity horizon of 10 days (large cap) and of 20 days (small cap, a further factor $\sqrt{20/10}$).
- ▶ (e) 95% moving-block bootstrap intervals (20-day blocks, 2000 resamples) for (a) and for the one-day stressed ES of (c), with the stress window held fixed.

□ Report

- ▶ A table with all numbers and the intervals of (e), and a one-page argument: which portfolio needs more capital, and how do thin trading and currency (RON vs USD) change the answer?

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive losses in %, and the 10-day VaR 1%.”

The assistant's answer

```
import numpy as np
r = 100 * np.log(px).diff().dropna()           # daily log returns, %
loss = -r                                       # loss in % of the position
var1 = np.percentile(r, 99)                   # VaR 1%
es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
var10 = 10 * var1                             # 10-day VaR 1%
```

- “VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46% of the position.”

C2: Critique an AI Answer [Proposed]: Task

Solution discussed in class

▣ Data and assumptions

- ▶ The same BET data as in B1 (2000–2026, daily log returns in %); the code on the previous slide runs without errors, but the answer contains **three** mistakes typical for VaR and ES; everything else is correct.

▣ Tasks

- ▶ (a) For each mistake, name the line and explain in one sentence why it is wrong, using the definitions $\text{VaR}_\alpha = -q_\alpha(r)$ and $\text{ES}_\alpha = \text{mean loss beyond VaR}_\alpha$.
- ▶ (b) Say how you would detect it: a count of observations, a number from B1, or a slide of the lecture.
- ▶ (c) Write the corrected line(s) and recompute the three numbers.
- ▶ Hint: count how many daily losses es25 averages, and how many it should average.

▣ Report

- ▶ Log each mistake as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, and the correct version. Model: B1.

Common Pitfalls

- ▣ Mixing sign conventions: a VaR reported as a negative return in one table and a positive loss in another
- ▣ Using the Normal distribution or Cornish–Fisher with daily kurtosis of 10
- ▣ Reporting ES 1% from 500 days without saying it averages 5 numbers
- ▣ Scaling by $\sqrt{10}$ a series with positive autocorrelation (BET, the BVB basket)
- ▣ Reading money weights as risk weights: a 10% crypto position can carry over a third of the risk
- ▣ Log returns summed across assets: portfolio losses need simple returns

Take-Home Assignment (Team Project)

- Choose a portfolio of at least three assets not covered in class (BVB stocks, a currency, a crypto-asset)
- Estimate one-day $\text{VaR}_{1\%}$ and $\text{ES}_{2.5\%}$ with historical, filtered and extreme-value methods, each with a bootstrap interval
- Split the ES into Euler contributions; from equal weights (long-only, fully invested, historical ES 2.5%, sample-mean expected return), minimise ES subject to the same expected return and aim for a 20% reduction; if it is infeasible, report the minimum attainable ES
- Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- Deliver a Quantlet folder (code, MetaInfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

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